D1.3 Mutation

NAME	DATE	TOTAL
		_____ / 36

About 35 minutes, or one section at a time. You will need a table of mRNA codons. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 DEFINE gene mutation. [D1.3.1] [1]

2 STATE three types of gene mutation. [D1.3.1] [1]

3 STATE the type of mutation that occurs in the HTT gene. [D1.3.3] [1]

4 LIST two chemical mutagens and two forms of mutagenic radiation. [D1.3.4] [2]

5 STATE two causes of gene mutation other than mutagens. [D1.3.4] [2]

Section B: Understanding

13 MARKS

6 EXPLAIN what is meant by a single-nucleotide polymorphism, and how one arises. [D1.3.2] [2]

7 EXPLAIN how the delta 32 mutation gives resistance to HIV. [D1.3.3] [3]

8 EXPLAIN what is meant by the randomness of mutation. [D1.3.5] [3]

9 COMPARE AND CONTRAST mutations in germ cells and in somatic cells. [D1.3.6] [3]

GERM CELLS	SOMATIC CELLS

10 EXPLAIN why meiosis cannot be the original source of new alleles. [D1.3.7] [2]

Section C: Working with sequences and data

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

11 [D1.3.2, D1.3.3]

A section of mRNA reads: A U G C C U G A A U G G

a DEDUCE the amino acid sequence. [1]

b The second codon changes to C C C. DEDUCE the effect, giving a reason. [2]

c The third codon changes to U A A. DEDUCE the effect. [1]

d The first C in the second codon is deleted. WRITE the new codons, and **deduce** the amino acid sequence. [2]

12 [D1.3.4, D1.3.5]

Barley seeds were given different doses of X-rays and then grown. The percentage of seedlings with visible changes was recorded.

X-RAY DOSE / ARBITRARY UNITS	0	10	20	40
SEEDLINGS WITH CHANGES / %	0.1	2.4	4.6	9.0

a DESCRIBE the relationship. [1]

b SUGGEST why some changes appeared at a dose of 0. [1]

c EXPLAIN why a researcher could not predict which gene would change in a given seed. [2]

Section D: Longer answer

6 MARKS

13 **EXPLAIN** the consequences of gene mutations, both for the individual and for a species. [D1.3.2-7] [6]